Sizhe Li

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EDUCATION

The Chinese University of Hong Kong, Shenzhen — *Shenzhen, China*
Bachelor of Science in Statistics — *Sep. 2022 – Present*
Cumulative GPA: 3.926/4.000 (rank: 1/48), Major GPA: 3.983/4.000

University of California, Berkeley — *California, USA*
Visiting Student — *Aug. 2024 – Dec. 2024*
GPA: 4.000/4.000

No.1 Middle School Affiliated to Central China Normal University — *Wuhan, China*
High School Diploma — *Sep. 2019 – Jun. 2022*

RESEARCH EXPERIENCE

Prediction-Specific Design of Learning-Augmented Algorithms — *Feb 2024 – May 2025*
Supervised by Prof. Tongxin Li and Prof. Nicolas Christianson

- **New Framework & Definitions:** Introduced a *prediction-specific framework* for learning-augmented algorithms enabling fine-grained, per-prediction performance guarantees. Proposed a novel definition of *strong optimality* capturing Pareto optimality in both consistency and robustness tradeoffs. Proved the suboptimality of state-of-the-art methods (NeurIPS 2018, 2021) in this new framework.
- **Generic Approach:** Developed a *bi-level optimization* framework for designing strongly-optimal algorithms, applicable to diverse online problems.
- **Explicit Algorithm Design:** Designed provably strongly optimal algorithms for *deterministic ski rental*, *randomized ski rental* and *one-max search*.

Optimal Tradeoff in Learning-Augmented Non-Clairvoyant Scheduling — *June 2025 – Present*
Supervised by Prof. Tongxin Li and Prof. Nicolas Christianson

- This work advances a central open problem in learning-augmented online scheduling: determining the optimal consistency–robustness trade-off for general n jobs. Prior bounds (Wei & Zhang, NeurIPS'20) were tight only for $n = 2$ and two extreme points. We prove that the classical lower bound is not tight for $n \geq 5$, establish refined lower and upper bounds for finite n along with the matching algorithm, and derive analytical lower and upper bounds in the asymptotic regime as $n \to \infty$, providing a near-optimal characterization of the frontier. A first-author preprint will be released soon.

PUBLICATIONS

- **Sizhe Li**, Nicolas Christianson, Tongxin Li. *Prediction-Specific Design of Learning-Augmented Algorithms*. arXiv:2510.14887, arXiv preprint. **ACM SIGMETRICS 2026**.

PROJECTS

ICU Discrete-Event Simulation (UC Berkeley) — *Aug. 2024 – Dec. 2024*
Built a discrete-event simulation based on the MIMIC-IV demo dataset to study ICU flow under capacity constraints. Implemented simple triage scoring and baseline allocation heuristics, and evaluated their behavior across varying demand regimes via stress testing and sensitivity analysis.

TEACHING

Undergraduate Teaching Fellow, CUHK-Shenzhen

- **MAT2041: Linear Algebra and Applications** Fall 2023, Spring 2025
- **CSC1001: Introduction to Computer Science: Programming Methodology** Fall 2025
- **STA4001: Stochastic Processes** Spring 2026

HONORS AND AWARDS

- Dean's List, 2023–2025
- Academic Performance Scholarship (Top 1–2%), 2023–2024
- Undergraduate Research Award, 2025

SKILLS

- **Algorithms and Theory:** Approximation algorithms, competitive analysis, complexity theory
- **Programming:** Python (NumPy, Pandas, SciPy, Matplotlib), Java, R, MATLAB
- **Tools:** Git, Linux, Jupyter, VS Code, Overleaf
- **Typesetting:** LaTeX